Core Pure 1 · Chapter 6

Matrices

Matrix notation · Arithmetic · Determinants · Inverses · Simultaneous equations

A **matrix** (plural: matrices) is an array of numbers arranged in rows and columns.

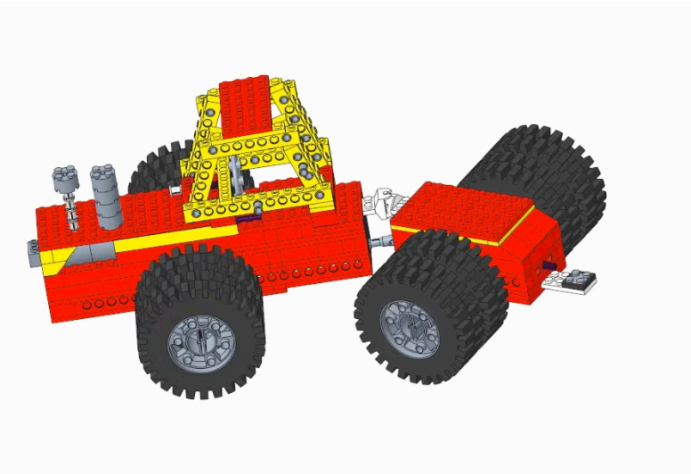
$$\begin{pmatrix} 1 & 0 & -2 \\ 3 & 3 & 0 \end{pmatrix}$$

A matrix organises several values into one structure.

Why use matrices?

They represent linear transformations, including rotations, reflections and enlargements, and help solve simultaneous equations.

In 3D graphics, matrices control transformations and camera angles.



Example of matrices 3

In computer science, matrices are also called arrays. This example records the top five teams in a Premier League table.

Columns: matches played, won, drawn, lost, goals for, goals against, goal difference, points.

$$\begin{pmatrix} 26 & 20 & 3 & 3 & 59 & 25 & 34 & 63 \\ 27 & 19 & 4 & 4 & 67 & 25 & 42 & 61 \\ 25 & 15 & 4 & 6 & 41 & 35 & 6 & 49 \\ 27 & 15 & 3 & 9 & 49 & 37 & 12 & 48 \\ 26 & 12 & 6 & 8 & 47 & 29 & 18 & 42 \end{pmatrix}$$

There are **5 rows** and **8 columns**: a 5×8 matrix.

Dimension of Matrices 4

The dimension of a matrix is its number of rows followed by its number of columns.

$$\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix}$$

1

$$\begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$$

2

$$\begin{pmatrix} 1 & 6 & 0 \end{pmatrix}$$

3

Matrix Notation 5

Matrices may use square or curved brackets. A matrix with one column is a column vector; a matrix with one row is a row vector.

$$\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 6 & 0 \end{pmatrix}$$

Matrix variables use **bold capital letters**; vector variables use bold lowercase letters.

$$\mathbf{A} = \begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$$

$$\mathbf{C} = \mathbf{P}^{-1}\mathbf{T}\mathbf{P}$$

Matrix Fundamentals: Adding/Subtracting Matrices 6

Arsenal and Liverpool draw 2–2. Add the match-result matrix to the original standings. Only matrices of the same dimensions can be added or subtracted.

Original standings

$$\begin{pmatrix} 26 & 20 & 3 & 3 & 59 & 25 & 34 & 63 \\ 27 & 19 & 4 & 4 & 67 & 25 & 42 & 61 \\ 25 & 15 & 4 & 6 & 41 & 35 & 6 & 49 \\ 27 & 15 & 3 & 9 & 49 & 37 & 12 & 48 \\ 26 & 12 & 6 & 8 & 47 & 29 & 18 & 42 \end{pmatrix}$$

Match result

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 2 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 2 & 2 & 0 & 1 \end{pmatrix}$$

Updated table

Now you try: 7

Add or subtract corresponding elements:

$$\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix} + \begin{pmatrix} 6 & -2 & 9 \\ 2 & 1 & 0 \end{pmatrix}$$

1

Add or subtract corresponding elements:

$$\begin{pmatrix} 3 & 0 \\ -1 & 2 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} q & -3 \\ 1 & 1 \\ -4 & 1 \end{pmatrix}$$

2

Scalar Multiplication 8

A scalar multiplies every element of a matrix, just as it does for a vector.

$$3\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix}$$

1

$$A = \begin{pmatrix} q & -3 \\ 1 & 1 \\ -4 & 1 \end{pmatrix}$$

Find $2A$.

2

Further Example using algebraic terms 9

Find the values of k such that

$$\begin{pmatrix} -3 \\ k \end{pmatrix} + k \begin{pmatrix} 2k \\ 2k \end{pmatrix} = \begin{pmatrix} k \\ 6 \end{pmatrix}$$

First row

Second row

Conclusion

Special Matrices 10

A **square matrix** has the same number of rows and columns.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 & 4 \\ 2 & 2 & 5 \\ -3 & 4 & 3 \end{pmatrix}$$

A **zero matrix** has every element zero.

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

An **identity matrix** has 1 on its leading diagonal and 0 elsewhere.

$$\mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Dimensions are usually clear from the context. Identity matrices are particularly useful in multiplication.

Practice: 11

1. Find a, b, c given

$$\begin{pmatrix} 1 & 2 & 0 \\ a & b & c \end{pmatrix} + \begin{pmatrix} a & b & c \\ 1 & 2 & 0 \end{pmatrix} = \begin{pmatrix} c & 5 & c \\ c & c & c \end{pmatrix}$$

1

2. Find a, b, c, d given

$$\begin{pmatrix} 5 & a \\ b & 0 \end{pmatrix} - 2\begin{pmatrix} c & 2 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 9 & 1 \\ 3 & d \end{pmatrix}$$

2

Multiply a Matrix by a vector 12

Coffi Lab sells espresso for £3, cappuccino for £4 and tea for £2.

Rows below: espresso, cappuccino, tea. Columns: Monday to Friday.

$$\begin{pmatrix} 13 & 9 & 7 & 15 & 1 \\ 8 & 7 & 4 & 6 & 2 \\ 6 & 4 & 0 & 3 & 5 \end{pmatrix}$$

How much did they make on Monday?

Monday

Find the revenue for each day.

Daily revenue

Multiply a Matrix by a vector 13

You have calculated a row-vector × matrix product.

$$\begin{pmatrix}3 & 4 & 2\end{pmatrix}\begin{pmatrix}13 & 9 & 7 & 15 & 1\\ 8 & 7 & 4 & 6 & 2\\ 6 & 4 & 0 & 3 & 5\end{pmatrix} = \begin{pmatrix}83 & 63 & 37 & 75 & 21\end{pmatrix}$$

Suppose production costs are 50% of selling prices. Form a matrix containing both prices and costs.

Prices and costs

Multiply to find daily revenue and daily cost.

Product

Matrix Multiplication 14

Multiply each row of the left matrix by each column of the right matrix, adding the products.

$$\begin{pmatrix}1 & 0 & 3 & -2\\ 2 & 8 & 4 & 3\\ 7 & -1 & 0 & 2\end{pmatrix}\begin{pmatrix}5 & 1\\ 1 & 7\\ 0 & 3\\ 8 & -3\end{pmatrix}$$

First row, first column

Rest of first column

Complete product

Matrix Multiplication involving I 15

What do you notice?

$$\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}3 & 2\\1 & 0\end{pmatrix}$$

Left identity

$$\begin{pmatrix}3 & 2\\1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$$

Right identity

For compatible identity matrices,

$\mathbf{AI} = \mathbf{IA} = \mathbf{A}$.

Multiplying by an identity matrix has no effect, just as multiplying a number by 1 has no effect.

The identity for ordinary addition is 0:

$a + 0 = 0 + a = a$.

Now you try.. 16

a. $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}\begin{pmatrix}2 & 0 & -1\\3 & 2 & 1\end{pmatrix}$

a

b. $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}\begin{pmatrix}3\\-1\end{pmatrix}$

b

c. $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}^2$

c

d. $\begin{pmatrix}1 & 2 & 3\end{pmatrix}\begin{pmatrix}1\\2\\3\end{pmatrix}$

d

e. $\begin{pmatrix}1\\2\\3\end{pmatrix}\begin{pmatrix}1 & 2 & 3\end{pmatrix}$

e

f. $\begin{pmatrix}1 & a\\0 & 1\end{pmatrix}^k$

f

Matrix Multiplication 17

The product AB exists only when the columns of A equal the rows of B.

$A_{m\times n}B_{n\times p} = (AB)_{m\times p}$

Only square matrices have positive integer powers greater than 1.

Find each product's dimensions, or state that it is invalid.

1. $2\times3 \times 3\times4$
2. $1\times3 \times 2\times3$
3. $6\times2 \times 2\times4$
4. $1\times3 \times 3\times1$

1

2

3

4

5. $7\times5 \times 7\times5$
6. $10\times10 \times 10\times9$
7. $3\times3 \times 3\times3$

5

6

7

Practice 18

3. Find in terms of a.

$$\begin{pmatrix}2 & a\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 3 & 0\\ 0 & -1 & 2\end{pmatrix}$$

3

4. Find in terms of x.

$$\begin{pmatrix}3 & 2\\ -1 & x\end{pmatrix}\begin{pmatrix}x & -2\\ 1 & 3\end{pmatrix}$$

4

5. Given

$$A = \begin{pmatrix}1 & 2\\ 0 & 1\end{pmatrix}$$

find A^2, A^3 and suggest A^k.

5

6. Given

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$$

prove that $A^2 = 2A + 5I$.

6

7. Given

$$A = \begin{pmatrix} p & 3 \\ 6 & p \end{pmatrix}, \quad B = \begin{pmatrix} q & 2 \\ 4 & q \end{pmatrix}$$

prove that $AB = BA$.

7

Matrix Multiplication on the CG50 20

Matrix multiplication on a graphing calculator

Enter Run-Matrix mode and define two 2×2 matrices.

$$A = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Enter each element in order. Return to the calculation screen, insert MatA and MatB using the matrix options menu, and evaluate MatA $\times$ MatB.

Button names vary by model.

Find AB, then check using the calculator.

Product

Matrix Multiplication on the Classwiz fx-991EX 21

Classwiz matrix practice

$$A = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix},\ B = \begin{pmatrix} 4 & 0 \\ 1 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 9 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix}$$

Open Matrix mode. Define MatA and MatB as 2×2, and MatC as 2×3. Enter each matrix. Use Matrix Calc and recall the stored matrices to evaluate the expressions.

Calculate

$4A,\ 3B - 2A,\ AB,\ BA,\ B^2,\ AC,\ BAC.$

a–e

f–g

Determinant of a matrix 22

A matrix can transform a point:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

Can an inverse transformation recover the original point?

$$\begin{pmatrix} ? & ? \\ ? & ? \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The required matrix is called the **inverse**.

Just as the discriminant $b^2 - 4ac$ distinguishes quadratic root cases, the **determinant**, written $\det A$ or $|A|$, determines whether a square matrix has an inverse.

Determinants 23

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\det A = |A| = ad - bc.$

If $\det A = 0$, A is singular and has no inverse. Otherwise it is non-singular.

Find each determinant.

1. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
2. $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
3. $\begin{pmatrix} 0 & 3 \\ -1 & -4 \end{pmatrix}$
4. $\begin{pmatrix} 10 & -2 \\ 4 & -1 \end{pmatrix}$

1

2

3

4

Test Your Understanding So Far 24

[Textbook] Given

$A = \begin{pmatrix} 4 & p+2 \\ -1 & 3-p \end{pmatrix}$

and that A is singular, find p.

Textbook

Edexcel FP1 January 2010 Q5

Let $a \in \mathbb{R}$ and

$A = \begin{pmatrix} a & -5 \\ 2 & a+4 \end{pmatrix}$

a. Find $\det A$.

b. Show that A is non-singular for all a.

a

b

Determinants of 3× 3 matrices 25

Expand along the first row, with signs $+, -, +$.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Example

$$\det \begin{pmatrix} 3 & 1 & 4 \\ 2 & 2 & 5 \\ -3 & 4 & 3 \end{pmatrix}$$

Determinant

Test Your Understanding 26

1. Determine the determinant.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 5 & -6 \\ -1 & 8 & 2 \end{pmatrix}$$

1

2. Given that the matrix is singular, find k.

$$A = \begin{pmatrix} 3 & k & 0 \\ -2 & 1 & 2 \\ 5 & 0 & k+3 \end{pmatrix}$$

2

Minors 27

A minor is the determinant left after deleting the row and column of an entry.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 5 & -6 \\ -1 & 8 & 2 \end{pmatrix}$$

Find the minor of

0

0

Find the minor of

-6

-6

Find the minor of

5

5

Exercise 6C Pure Core 1 - P106-108 28

2a. Find the value of a for which the matrix is singular.

$$\begin{pmatrix} a & 1+a \\ 3 & 2 \end{pmatrix}$$

a

2b. Find the value of a for which the matrix is singular.

$$\begin{pmatrix} 1+a & 3-a \\ a+2 & 1-a \end{pmatrix}$$

b

2c. Find the value of a for which the matrix is singular.

$$\begin{pmatrix} 2+a & 1-a \\ 1-a & a \end{pmatrix}$$

c

3. Find the exact real values of k for which

$$M = \begin{pmatrix} -2 & 1-k \\ k-1 & k \end{pmatrix}$$

is singular.

Answer

4. Find the values of k for which

$$P = \begin{pmatrix} 3k & 4-k \\ k-2 & -k \end{pmatrix}$$

is singular.

Answer

8. Given

$$A = \begin{pmatrix} 2 & 1 & -4 \\ 2k+1 & 3 & k \\ 1 & 0 & 1 \end{pmatrix}$$

is singular, find k.

8

11. Show that the matrix is singular for all a, b, c.

$$\begin{pmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{pmatrix}$$

11

13. Find the determinant of

$$A = \begin{pmatrix} x-3 & -2 & 0 \\ 1 & x & -2 \\ -2 & -1 & x+1 \end{pmatrix}$$

and hence the values of x for which it is singular.

Determinant

Singular values

14. Given

$$M = \begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} -1 & k \\ 4 & 3 \end{pmatrix}$$

a. Find $\det M$.

b. Given $\det N = \det M$, find k.

a

b

c. Find MN.

d. Verify that $\det(MN) = \det M \det N$.

c

d

Inverting a 2×2 matrix 34

The inverse satisfies $AA^{-1} = A^{-1}A = I$.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

This exists only when $\det A \neq 0$.

To reverse $A\mathbf{x} = \mathbf{y}$, multiply on the left by A^{-1}:

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{y}$$

$$\mathbf{x} = A^{-1}\mathbf{y}$$

Practising the Inverse 35

Find the inverse, step by step.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

First check that $ad - bc \neq 0$.

1: Swap

2: Negate

3: Divide

Test Your Understanding 36

a. Find the inverse of

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

a

b. Find the inverse of

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

b

c. Find the inverse of

$$\begin{pmatrix} 7 & 2 \\ 1 & -3 \end{pmatrix}$$

c

d. Find the inverse of

$$\begin{pmatrix} 4 & p+2 \\ -1 & 3-p \end{pmatrix}$$

d

Matrix Proofs Involving Inverse

For non-singular matrices P, Q, show that $(PQ)^{-1} = Q^{-1}P^{-1}$.

1

Given $BAB = I$, prove that $A = B^{-1}B^{-1}$.

2

Exercise 6D Pure Core 1 - P110-111

3a. For invertible square matrices, $ABC = I$. Prove that $B^{-1} = CA$.

a

3b. Hence find B when

$$A = \begin{pmatrix} 0 & 1 \\ -1 & -6 \end{pmatrix}$$

$$C = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix}$$

b

7. Find A given

$$B = \begin{pmatrix} 5 & -4 \\ 2 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 11 & -1 \\ -8 & 9 \\ -2 & -1 \end{pmatrix}$$

Inverse

A

8. Let $a, b \neq 0$ and

$$A = \begin{pmatrix} 3a & b \\ 4a & 2b \end{pmatrix}$$

a. Find A^{-1}.

b. Given

$$B = \begin{pmatrix} -a & b \\ 3a & 2b \end{pmatrix}$$

and $B = XA$, find X.

a

b

10. Let

$$M = \begin{pmatrix} 2 & 3 \\ k & -1 \end{pmatrix}$$

a. For which k does the inverse exist?

b. Find the inverse.

a

b

13. Let

$$A = \begin{pmatrix} a & 2 \\ 3 & 2a \end{pmatrix}$$

a. Find A^{-1}.

b. Find the two values of a for which no inverse exists.

a

b

Proof for the Formula of the Inverse of a 2x2 Matrix 43

For interest: derive the inverse by row operations. Let $\Delta = ad - bc \neq 0$. For $a \neq 0$, start with

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right]$$

Divide row 1 by a, then subtract c times row 1 from row 2:

$$\left[\begin{array}{cc|cc} 1 & b/a & 1/a & 0 \\ 0 & \Delta/a & -c/a & 1 \end{array}\right]$$

Multiply row 2 by a/Δ, then subtract b/a times row 2 from row 1:

$$\left[\begin{array}{cc|cc} 1 & 0 & d/\Delta & -b/\Delta \\ 0 & 1 & -c/\Delta & a/\Delta \end{array}\right]$$

$$M^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

If $a = 0$, interchange the rows first. The same formula also follows by multiplying the proposed inverse by M.

Matrix Transpose 44

The transpose A^T interchanges rows and columns.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

An $m \times n$ matrix becomes $n \times m$.

Beyond the exam: transposition moves a transformation between the two vectors in a dot product.

$$(A\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^T\mathbf{v})$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

$$\begin{pmatrix} 17 \\ 39 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 8 \end{pmatrix} = 431$$

$$\begin{pmatrix} 5 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 31 \\ 46 \end{pmatrix} = 431$$

Inversing a 3×3 matrix 45

General inverse method, illustrated with

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

1. Find the determinant.
2. Find each minor by deleting its row and column.
3. Apply alternating signs, beginning with $+$ at the top left, to form the cofactor matrix C.
4. Transpose C and divide by the determinant.

1–2

3–4

Inversing a 3×3 matrix 46

Find the inverse of

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 4 & 1 \\ 2 & -1 & 0 \end{pmatrix}$$

1: Determinant

2: Minors

3: Cofactors

4: Inverse

Doing with your Classwiz 47

Use your Classwiz to invert

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 4 & 1 \\ 2 & -1 & 0 \end{pmatrix}$$

1. Open Matrix mode and select MatA.
2. Choose 3 rows and 3 columns, then enter the numbers, pressing $=$ after each.
3. Press AC to start a calculation.
4. Use OPTN to insert MatA.
5. Use the dedicated x^{-1} key, not the general power key.
6. $Press =$.

Menu names depend on the calculator model.

Further Example 48

Given

$$A = \begin{pmatrix} -2 & 3 & -3 \\ 0 & 1 & 0 \\ 1 & -1 & 2 \end{pmatrix}$$

$$(AB)^{-1} = \begin{pmatrix} 8 & -17 & 9 \\ -5 & 10 & -6 \\ -3 & 5 & -4 \end{pmatrix}$$

a. Show $A^{-1} = A$.

b. Find B^{-1}.

a

b

June 2011 Q7:

$$M = \begin{pmatrix} k & -1 & 1 \\ 1 & 0 & -1 \\ 3 & -2 & 1 \end{pmatrix}$$

where $k \neq 1$.

a. Show $\det M = 2 - 2k$.

b. Find M^{-1}.

a

b

1. Use your calculator to find the inverse; the cofactor method is shown for comparison.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Determinant and minors

Cofactors

Inverse

4. Let

$$A = \begin{pmatrix} 2 & 0 & 3 \\ k & 1 & 1 \\ 1 & 1 & 4 \end{pmatrix}$$

a. Show $\det A = 3(k+1)$.

b. For $k \neq -1$, find A^{-1}.

a

b: Minors and cofactors

b: Inverse

5. Given

$$A = \begin{pmatrix} 5 & a & 4 \\ b & -7 & 8 \\ 2 & -2 & c \end{pmatrix}$$

and $A = A^{-1}$, find a, b, c.

Constants

9. Let

$$M = \begin{pmatrix} 2 & k & 3 \\ -1 & 2 & 1 \\ 1 & -1 & -1 \end{pmatrix}$$

a. For which k does M^{-1} exist?

b. Find M^{-1}.

a

b: Cofactors

b: Inverse

10. Given that

$$A = \begin{pmatrix} p & 2p & 3 \\ 4 & -1 & 1 \\ 1 & -2 & 0 \end{pmatrix}$$

is non-singular, find A^{-1}.

Determinant

Cofactors

Inverse

Using Matrices For Simultaneous Equations 55

Solve the simultaneous equations using an inverse matrix.

$$-x + 6y - 2z = 21$$
$$6x - 2y - z = -16$$
$$-2x + 3y + 5z = 24$$

1: Matrix equation

2: Inverse

3: Solution

Forming the equations yourself 56

A population of 1000 mole-rats has 100 more females than males. After one year, the numbers of males and females rise by 2% and 3% respectively; the number of young falls by 4%. The total population falls by 20. Find the initial numbers.

1: Model

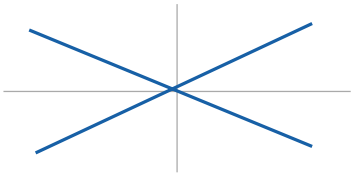
2: Invert

3: Population

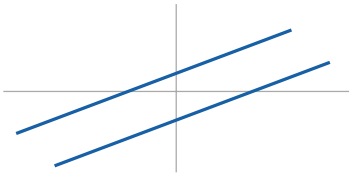
Consistency of linear equations 57

A system is consistent if it has at least one solution. In two dimensions, solutions are common points of the lines.

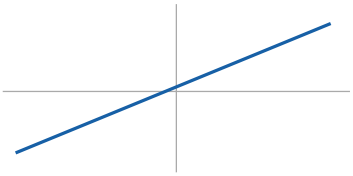
1. One intersection: unique solution.
   $x + 2y = 1$
   $3x - y = 0$



2. Distinct parallel lines: no solution.
   $x - 3y = 1$
   $x - 3y = 2$



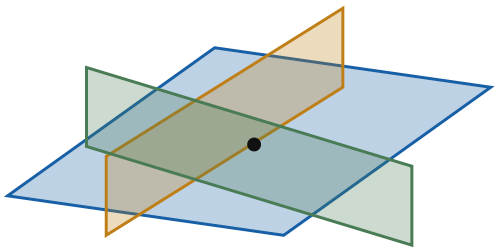
3. Coincident lines: infinitely many solutions.
   $x - 3y = 1$
   $2x - 6y = 2$



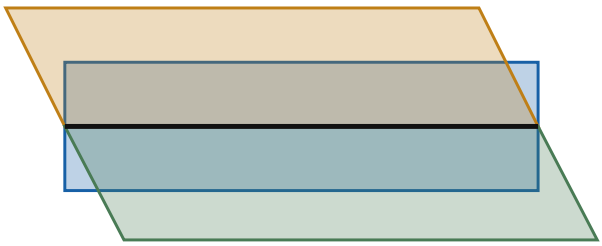
Extending consistency to 3 variables 58

In three dimensions, $ax + by + cz = d$ describes a plane. A simultaneous solution belongs to all three planes.

1. One common point: unique solution.



2. One common line (a sheaf): infinitely many solutions.



3. Prism arrangement: pairwise intersections, but no common point. Inconsistent.



4. Distinct parallel planes: no common point. Inconsistent.



Extending consistency to 3 variables 59

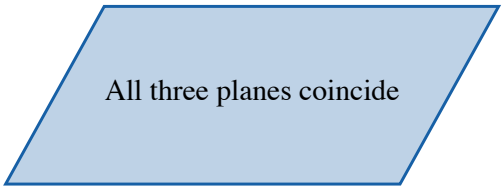
5. Coincident planes

If all three equations describe the same plane, every point on that plane is a solution.

$x + 2y - z = 3$

$2x + 4y - 2z = 6$

$-x - 2y + z = -3$



All three equations are multiples of one another.

The system is consistent with infinitely many solutions.

A zero determinant alone does not tell us whether there are no solutions or infinitely many: we must inspect the equations.

Example 60

Describe the intersections when $k = 0, 1, -6$.

$3x - ky - 6z = k$

$kx + 3y + 3z = 2$

$-3x - y + 3z = -2$

First take $k = 0$.

Determinant

Intersection

Example 61

For $k = 1$:

$$3x - y - 6z = 1$$
$$x + 3y + 3z = 2$$
$$-3x - y + 3z = -2$$

k = 1

For $k = -6$:

$$3x + 6y - 6z = -6$$
$$-6x + 3y + 3z = 2$$
$$-3x - y + 3z = -2$$

k = −6

Test Your Understanding 62

Find the values of k for which the system has a unique solution.

$$2x + 3y - z = 13$$
$$3x - y + kz = 11$$
$$x + y + z = 7$$

Determinant

Condition

2. Solve the system and show that it has a unique solution.

$$x - 3y - 4z = 3$$
$$6x + 5y - 7z = 30$$
$$x + 4y + 6z = -3$$

Matrix

Inverse

Solution

3. Phyllis invests £3000 in funds A, B and C, putting £190 more in A than B. After two years their values change by $+1\%$, $+2.5\%$ and -1.5% respectively. Her total investment increases by £41. Find the original investments.

Model

Inverse

Answer

5. These planes do not intersect at a single point. Find a and describe their intersection.

$$x + ay + 2z = a$$
$$x - y - z = a$$
$$x + 4y + 4z = 0$$

Parameter

Intersection

6. Given

$$M = \begin{pmatrix} 1 & 4 & q \\ 2 & 3 & -3 \\ q & q & -2 \end{pmatrix}$$

a. Show that singularity implies $q^2 + 9q - 10 = 0$.

b. Classify the system $M\mathbf{x} = (-16, q/2, -2)^T$ for $q = -10, 2, 1$.

a

b: q = −10

b: q = 2

b: q = 1